UniQuery: Unifying Vision Models with a Joint Query Head for Cross-Domain Pedestrian Attribute Recognition

Anonymous WACV Algorithms Track submission

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Abstract

In this paper, we present our solution to Track 1, Pedestrian Attribute Recognition (PAR), of the UPAR Challenge 2027 at the WACV 2027 Real-World Surveillance workshop. PAR provides semantic descriptions of individuals, enabling attribute-based person retrieval and supporting person re-identification. However, accurately recognizing fine-grained attributes across domains remains challenging due to occlusion, appearance variations, and shifts in image distributions. Ensembles of independently trained foundation encoders combine their logits with fixed fusion weights, so every attribute uses the same mix and the encoders are never trained together. Our model, UniQuery, instead jointly fine-tunes DINOv3 [28] and CLIP [23] for attribute-specific fusion. It combines two query heads, one for each encoder, with a joint query head whose attribute queries attend to tokens from both encoders, and the prediction is the average of the logits of the three query heads. Trained on PA100k [21], PETA [8], and Market-1501 [40], the model outperforms the independent ensemble in Challenge_Avg on both evaluated leave-one-dataset-out splits. Ablations show that learning fusion over independently trained encoders does not improve upon logit averaging, while freezing the encoders lowers the Challenge_Avg of UniQuery. On the hidden challenge test set, our approach achieves a Challenge_Avg of 0.8001.

1. Introduction

Pedestrian analysis plays an important role in intelligent video surveillance and online retail applications. A key task, Pedestrian Attribute Recognition (PAR), predicts semantic attributes of individuals from images, providing interpretable descriptions that support attribute-based person retrieval and person re-identification. Despite recent progress, robust recognition remains challenging due to occlusion, low image resolution, label imbalance, and variations in appearance and imaging conditions [18, 29, 41]. Recognition becomes harder still on a camera that was never seen in training, because resolution, viewpoint and lighting change, and each training dataset labels attributes by its own rules [29].

Foundation encoders pretrained on large and varied image collections [23, 28] are a natural starting point for this shift. They are usually adapted to downstream tasks by fine-tuning one encoder with a small head, or by averaging the logits of several independently fine-tuned encoders [16]. Earlier UPAR entries followed these routes, with a single fine-tuned Swin-B [6], thresholds tuned for each attribute on validation data [7], or two encoders merged into one vector shared by all attributes [2]. An average of independently fine-tuned encoders uses the same mix for every attribute, and its encoders are never trained together.

In this paper, we present our approach to Track 1 of the UPAR Challenge 2027. Our model, UniQuery (Fig. 1), jointly fine-tunes DINOv3 and CLIP, and each encoder has its own query head. Alongside these two query heads, a joint query head combines features from both encoders through one query per attribute, so that each attribute can draw different information from the two encoders. At inference, we average the logits of the three query heads. To evaluate rigorously how effective the method is, we build an evaluation protocol in which every design choice is tested on a dataset held out from training. The main contributions of this paper are below.

- We propose UniQuery, which fine-tunes DINOv3 and CLIP together and lets one query per attribute decide how to mix the two encoders. It outperforms an ensemble of independently fine-tuned encoders trained with the same loss on both held-out datasets, and it is also above the ensemble on the hidden test set, by a small margin. The gain needs the joint query head, because when it is removed at inference, the average of the other two query heads does not beat the ensemble.
- We show that, in our experiments, the mix needs to be learned together with fine-tuning the encoders. Fusion heads trained on top of independently fine-tuned encoders

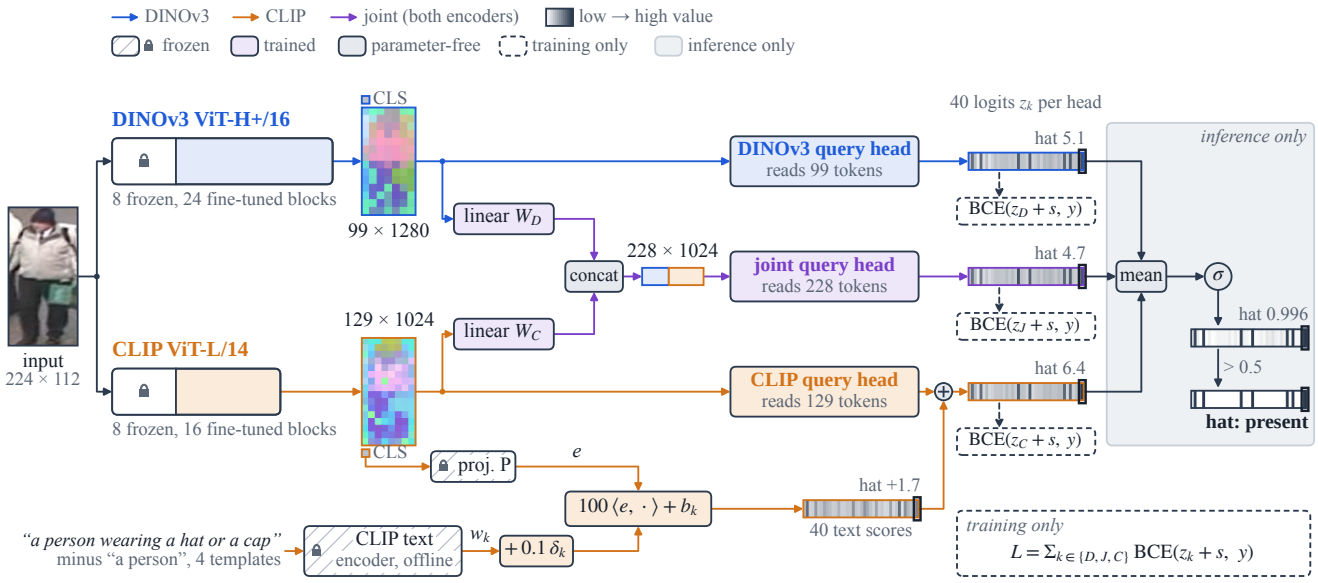


Figure 1. UniQuery on one PETA image. Striped parts with a lock are frozen, light tints mark trained parts, gray boxes are operations without parameters, and token cells are colored by the PCA of the patch tokens. In the text-score branch, P is the pretrained CLIP projection, e the projected and normalized CLS token, w_k the vector of the sentence for attribute k minus that of `a person`, and δ_k and b_k are learned. In training, each query head has its own loss with the shift s (Eq. (1)). At inference, the logits of the three query heads are averaged and compared with a threshold of 0.5. The submitted model saw this image in training.

do not beat a plain logit average in the mean over the two held-out datasets, and with frozen encoders UniQuery falls below a single fine-tuned DINOv3.

- We analyze how the joint query head mixes the encoders. Each of its attention heads reads almost only the tokens of one encoder, and the number of attention heads given to DINOv3 or CLIP changes with the attribute. For most attributes, the joint query head is more accurate than the average of the two single-encoder query heads, and for attributes other than colors it often matches or beats the better one.

2. Related work

Cross-domain and unified learning have attracted researchers because high accuracy on an individual PAR dataset does not guarantee reliable performance in a new environment [13, 29]. At the representation level, ASRL [39] reduces the classifier confusion caused by attribute correlations, by using one feature per attribute and a fixed orthogonal classifier. AttriVision [27] fine-tunes both CLIP encoders so that an image embedding is close to the embeddings of short sentences describing its attribute values, and predicts each attribute by cosine similarity. Like AttriVision, UniQuery uses CLIP text embeddings, but it keeps the text encoder frozen. It only adds a text score to the logits of the CLIP query head, and this score comes from the similarity of the image to attribute sentences.

A related direction uses images of the target domain. AdaGPAR [19] keeps its weights fixed at test time and classifies new images by their similarity to prototypes built from reliable target predictions. LDCD-PAR [42] also trains on unlabeled target images, with an adversarial domain discriminator on local features. In contrast, UniQuery is a domain generalization method and uses no target-domain image.

Unified learning offers another route to transferable representations. UniHCP [5] trains one ViT encoder and one decoder for many human-centric tasks, where each task has its own queries and PAR has one query per attribute. The PATH method of HumanBench [31] pretrains one shared encoder on many human-centric datasets with one projector per task. CLEAR [3] reuses its attribute recognition network for person retrieval by passing a sentence that describes the query attributes through a GPT language model and training only small adapters. Like UniHCP and query-based multi-label classifiers [20, 26], UniQuery uses learned queries that read the image tokens, but the queries of its joint query head read the tokens of two different encoders.

The two encoders of UniQuery differ in how they are pretrained. DINOv3 [28] learns by self-supervised learning on unlabeled images and gives high-quality patch features, while CLIP [23] and its successors [1, 11, 30, 34, 38] learn to match images with captions on web data. The variety of this web data is the main source of the robustness of CLIP

to distribution shift [10]. Still, CLIP treats some clearly different images as similar, while a self-supervised encoder separates them [33]. Adding self-supervised features next to CLIP improves multimodal language models on vision-centric benchmarks [32, 33], so we pair DINOv3 with CLIP.

A simple way to combine two encoders is an ensemble. Each encoder is fine-tuned alone into its own model, and the predictions of the two models are averaged at inference [16]. An ensemble only helps on images where the two models predict differently [15]. It also uses the same mix for every attribute. Weight averaging [36, 37] cannot merge DINOv3 with CLIP, because it needs models with the same starting point. Distillation merges several encoders into one [24], but the student is still worse than its teachers on some tasks, such as zero-shot classification against its image–text teacher [24, 25]. Stacking learns the mix on models that are already trained [35]. In UPAR 2024 [7], C^{2}T-Net [2] trained a Swin-B and an EVA ViT-L together and merged them into one vector for all attributes. UniQuery also trains the two encoders together, but each attribute has its own query that sets the mix, and each encoder keeps its own query head.

Attributes also differ in frequency, and most of the 40 attributes have few positive samples. A model trained in the usual way is biased toward predicting such attributes as absent [22]. This bias lowers the mean balanced accuracy (mA), which averages over attributes the accuracy on positives and the accuracy on negatives with equal weight. It therefore also lowers Challenge_Avg, the challenge metric, which is the harmonic mean of mA and instance F1. DeepMAR [17] corrects for this by weighting the loss of each attribute by its positive ratio, and UniQuery also uses these weights. Earlier UPAR entries also tuned one threshold per attribute on validation images of the source domains [7], so these thresholds may not hold on new cameras. Logit adjustment [22] adds a shift based on the attribute frequency to the logits inside the training loss, so UniQuery keeps one threshold of 0.5 for all attributes at inference.

3. The proposed method

Figure 1 shows the architecture of UniQuery. A cropped pedestrian image of 224 × 112 pixels passes through two encoders, DINOv3 ViT-H+/16 [28] and the OpenAI CLIP ViT-L/14 image encoder [23], which are fine-tuned in one network. Three query heads read the tokens of the two encoders in three ways. The DINOv3 and CLIP query heads read only the tokens of their own encoder, and the joint query head reads the tokens of both. Each query head gives 40 logits, one per binary attribute. During training, each query head has its own loss, and at inference the logits of the three query heads are averaged.

The CLIP query head also uses the image–text matching that CLIP learned in pretraining. For the hat attribute,

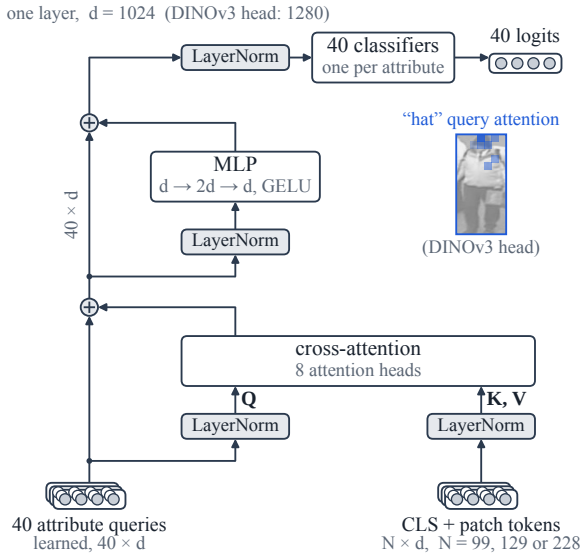


Figure 2. One query head. The three query heads of UniQuery share this design and differ only in the tokens they read. The inset is the attention of the hat query in the DINOv3 query head on the image of Fig. 1, averaged over the eight attention heads.

CLIP compares the image with the sentence `a person wearing a hat or a cap` and with the sentence `a person`. The text score of this attribute is the similarity of the image to the first sentence minus its similarity to the second (Fig. 3). This score is added to the hat logit that the CLIP query head computes from the image tokens. The subtraction aims to remove the similarity to `a person` that every attribute sentence shares. The addition aims to bring the knowledge of CLIP about the attribute into the logit computed from the image tokens.

The tokens of DINOv3 and CLIP have different widths, so before the joint query head each kind of token passes through its own linear layer to reach a common width. The tokens of the two encoders are then concatenated into one sequence that the joint query head reads. Inside, each query head consists of a cross-attention layer, a multilayer perceptron (MLP) and a classifier [20, 26] (Fig. 2).

During training, the loss in Eq. (1) is applied to all three query heads:

$$L = \sum_{k\in\{D,J,C\}} \sum_{a=1}^{40} m_a \, w_a(y_a) \, \mathrm{BCE}\big(z_{k,a} + s_a,\ y_a\big), \quad (1)$$

in which $s_a = \tau \log \frac{p_a}{1-p_a}$, p_a is the frequency of attribute a in the training data, and $\tau = 0.32$. The shift s_a applies logit adjustment [22] and is used only in training. The DeepMAR weight $w_a(y_a)$ [17] is e^{1-p_a} for positives and e^{p_a} for negatives, which moves the decision threshold by $1 - 2p_a$ on the logit scale. The loss therefore shifts the decision in two parts, $1 - 2p_a$ from w_a and a part from s_a.

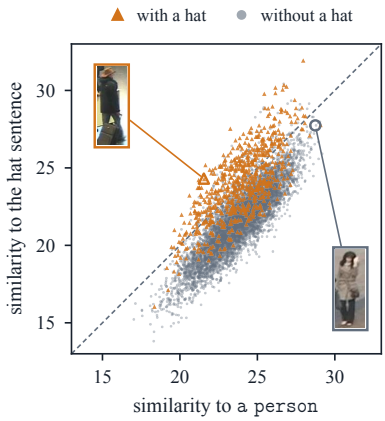


Figure 3. The idea of the text score for the hat attribute, on PETA validation images. The text score of an image is the vertical distance from its point to the diagonal. The image without a hat is more similar to the hat sentence than the image with a hat, but it lies below the diagonal and so gets a lower text score.

We chose $\tau = 0.32$ so that their sum fits, by least squares over the 40 attributes, the total shift of a post-hoc rule. In this rule, a model trained with the DeepMAR weights alone adds $0.42\left(\log\frac{1-p_a}{p_a} - (1-2p_a)\right)$ to its logits after training. The coefficient 0.42 of this rule was chosen on the leave-one-dataset-out runs. We denote by D the DINOv3 query head, J the joint query head, and C the CLIP query head. Here, $z_{k,a}$ denotes the logit predicted by query head k for attribute a, and y_a is the target label after applying label smoothing with a factor of 0.05. The mask m_a excludes attributes that are not annotated in the dataset to which the input image belongs.

During training, we freeze the patch embedding and the first eight blocks of each encoder and fine-tune the rest with AdamW for three epochs. We use a batch size of 64 and an exponential moving average (EMA) of the weights. The learning rate is 5×10^{-5} at the top block, decays by 0.75 per block below, and is ten times higher in the query heads. The submitted model is trained on the training and validation splits of all three datasets. At inference, the average logit of the three query heads passes through a sigmoid, and an attribute is predicted as present when the probability exceeds 0.5. Like an ensemble of the two separately trained encoders, UniQuery needs only two encoder forward passes per image. Its measured latency is about 3.0 ms per image on an NVIDIA RTX 5090 GPU with batches of 128 images.

4. Experiments

We evaluated independently trained models, each query head of UniQuery, and UniQuery with its three query heads averaged. The average of the three query heads reaches a higher mean Challenge_Avg than each query head alone and than the independent ensemble, so we use it for our final submission.

Model	PETA	Market	Avg.
Independently trained DINOv3	81.46	86.31	83.89
Independently trained CLIP	81.44	85.50	83.47
Ensemble of the two encoders, seed 1	82.45	86.30	84.38
Ensemble of the two encoders, seed 2	82.29	86.33	84.31
UniQuery, DINOv3 query head only	81.26	86.07	83.67
UniQuery, CLIP query head only	81.45	85.56	83.51
UniQuery, joint query head only	82.26	86.27	84.26
UniQuery, DINOv3 + CLIP query heads	82.24	86.32	84.28
UniQuery, three query heads averaged	**82.54**	86.39	84.46
Variant, seed 1	82.53	**86.44**	**84.49**
Variant, seed 2	82.51	86.32	84.42
Variant with two joint query heads	82.51	86.43	84.47

Table 1. Held-out Challenge_Avg with the same loss as in Eq. (1). Avg. is the mean of the two folds, and the submitted setting is UniQuery with its three query heads averaged. The row "DINOv3 + CLIP query heads" averages these two query heads of the same model, without the joint query head. The variant adds a text score for clothing colors on the upper or lower body region, with a different set of sentences, and in training hides each encoder from the joint query head with probability 0.25.

Protocol. To mimic the unseen cameras of the hidden test set, we use a leave-one-dataset-out protocol. In each fold, one dataset is held out completely, and the model sees none of its images during training. The model is trained on the training splits of the other two datasets and scored on the validation split of the held-out dataset. The PETA fold trains on PA100k and Market-1501 and is scored on the 6,963 PETA validation images. The Market fold trains on PA100k and PETA and is scored on the 16,458 Market-1501 validation images. We do not use the fold that holds out PA100k, because only 18.7k training images would remain. The metric is Challenge_Avg, in percent. Fold models are trained like the submitted model, except that their training data exclude the validation splits. Each image passes through the model once, without flipping, as in the submission. Results are for one seed unless stated otherwise. The factor 0.42 used to set τ and the encoder paired with DINOv3 were chosen on these same folds, so the fold scores may be slightly optimistic for these choices. The hidden test set is the only evaluation that these choices do not affect.

UniQuery versus the independent ensemble. Table 1 compares UniQuery with an ensemble of the two encoders trained independently, with the same encoder architectures and the same loss. The logit average of the three query heads of UniQuery beats both seeds of the ensemble on

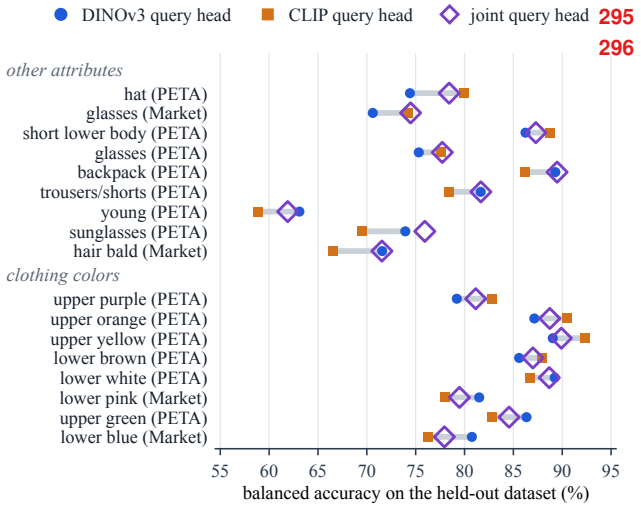


Figure 4. The joint query head next to the two single-encoder query heads, for the attributes on which these two differ by at least 2 points and that have at least 50 positive images. Values are means over two seeds of the variant in Tab. 1.

both folds. This gain needs the joint query head. When the joint query head is removed at inference, the average of the DINOv3 and CLIP query heads of UniQuery is no longer above the ensemble. Used alone, the joint query head is above each single-encoder query head but it does not beat the ensemble either. The gain appears only when the joint query head is averaged with the two single-encoder query heads.

Consistency. UniQuery beats the ensemble by only 0.08 to 0.15 points in the mean, close to the 0.07-point gap between the two seeds of the ensemble. We therefore check whether the gain repeats in other training runs. The last three rows of Tab. 1 are three runs of a UniQuery variant with a few small changes, given in the table caption, not reruns of UniQuery with other seeds. All four runs, UniQuery and the three runs of the variant, have a two-fold mean above both seeds of the ensemble. Fold by fold, the four runs win all 8 comparisons with the two ensemble seeds on PETA and 7 of 8 on Market.

Attribute-specific mixing. The joint query head does not mix the two encoders with one ratio for all attributes. Figure 4 compares it with the two single-encoder query heads of the UniQuery variant in Tab. 1. Over all attributes, both seeds and both folds, the joint query head is above the mean of the two single-encoder query heads in 82% of cases. For attributes other than colors, such as age, gender, hair, clothing type and accessories, it matches or beats the better single-encoder query head in 69% of cases, for example for backpack, sunglasses and trousers/shorts. For clothing colors, this happens in only 40% of cases, and the joint query head often lies between the two. Its attention shows that it shares the work between the encoders by attribute (Fig. 5). On the held-out PETA images, in 93% of the cases where an attention head reads an image for one attribute, more than 98% of its attention falls on the tokens of a single encoder (Fig. 5b). We can therefore count, for each attribute, how many of the eight attention heads read DINOv3 (Fig. 5c). An attention head reads DINOv3 if more than half of its attention falls on DINOv3 tokens. Backpack uses 6.4 of the eight attention heads on DINOv3 on average, while red and pink upper clothing use only about 2.6. In general, lower-body and accessory attributes rely more on DINOv3, and upper-body colors rely more on CLIP.

Why DINOv3 and CLIP? Table 2 shows that the second encoder should differ from the first. A second fine-tuned copy of the same encoder adds at most 0.10 points in the mean, while a different encoder adds 0.4 to 0.6 points. This gain comes from PETA. On Market, adding CLIP helps much less, and with $\tau = 0.42$ the pair of DINOv3 and CLIP is even below DINOv3 alone. Our hypothesis is that two encoders pretrained in different ways are wrong on the same image less often than two fine-tuned copies of one encoder. To test this, we take the image–attribute pairs that one model gets wrong, measure the share that the other model also gets wrong, and average over both directions. Two fine-tuned copies of the same encoder in Tab. 2 share 89% to 93% of their errors, while DINOv3 and CLIP share only 72% on PETA and 77% on Market-1501. To choose the encoder next to DINOv3, we tried 12 encoders on the PETA fold: CLIP, AIMv2 [12], EVA02-CLIP [30], MetaCLIP2 [4], MobileCLIP2 [9], DFN-CLIP [11], two PE models [1] and four SigLIP2 models [34]. Each is fine-tuned alone and its logits are averaged with those of DINOv3. Every one of them shares only 66% to 72% of its errors with DINOv3, so the Challenge_Avg of the pair follows that of the second encoder alone. Alone, CLIP is at least 0.5 points above every other encoder, and over five CLIP runs the pair of DINOv3 and CLIP reaches 81.04% to 81.23%. Only the pair with AIMv2-L falls in this range, and the other 10 encoders give lower pairs. The training recipe was tuned on DINOv3 and CLIP, so it may suit the other encoders less well. Inside UniQuery, replacing CLIP with SigLIP2-L/16 or C-RADIOv4-H [25], or replacing DINOv3 with C-RADIOv4-H, lowers the two-fold mean of Challenge_Avg (Tab. 4). The drop comes mostly from PETA. On Market, the pair with SigLIP2-L is also lower, while the two pairs with C-RADIOv4-H are slightly above UniQuery.

Can the mix be learned after training? The independent ensemble mixes its two models by averaging their logits, with the same equal ratio for every attribute. We test

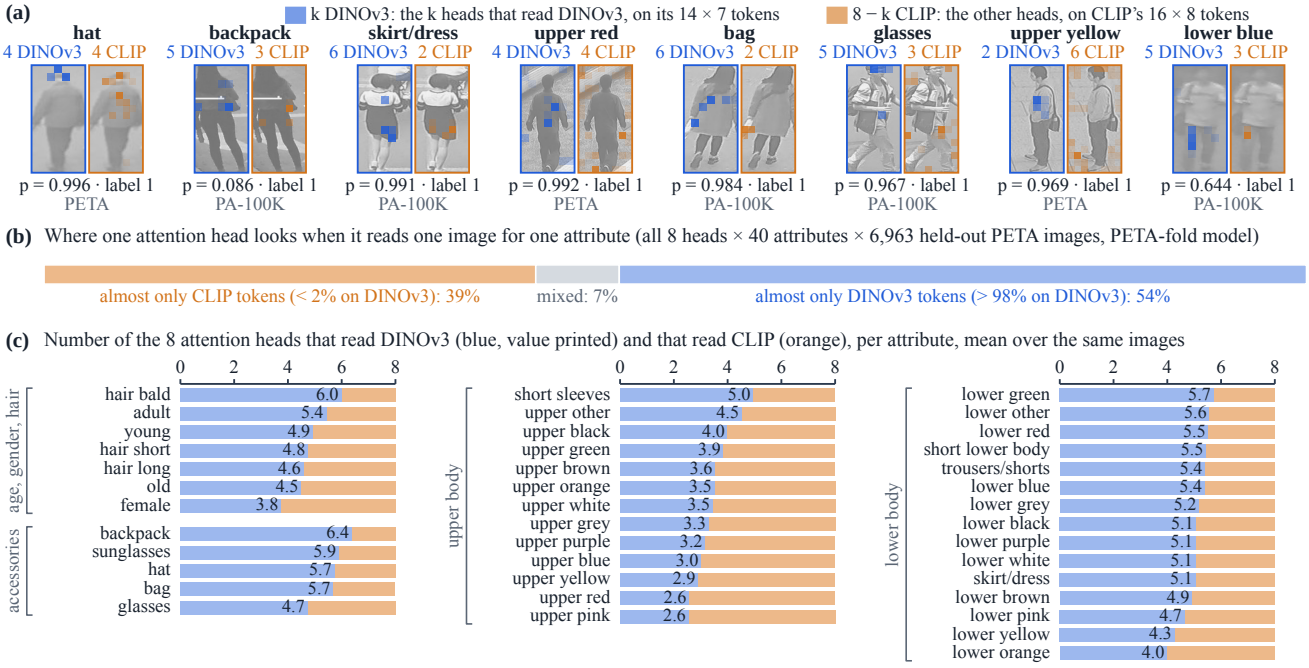


Figure 5. How the joint query head uses the two encoders. (a) Attention of the attention heads that read DINOv3 and of those that read CLIP, on eight validation images that the submitted model saw in training. (b, c) Statistics over the 6,963 held-out PETA images, for the PETA-fold model.

Encoders (logits averaged)	PETA	Market	Avg.
Logit-adjusted loss, $\tau = 0.42$			
CLIP	81.79	85.72	83.75
CLIP + CLIP (two seeds)	81.80	85.78	83.79
DINOv3	81.14	**86.78**	83.96
DINOv3 + CLIP	**82.29**	86.49	**84.39**
Weighted BCE			
CLIP	80.68	85.01	82.84
CLIP + CLIP (two seeds)	80.69	85.12	82.90
DINOv3	80.49	85.31	82.90
DINOv3 + DINOv3 (two runs)	80.66	85.35	83.00
DINOv3 + CLIP	**81.08**	**85.54**	**83.31**

Table 2. Held-out Challenge_Avg of separately fine-tuned encoders with averaged logits. These runs come from an earlier stage, trained with $\tau = 0.42$ instead of 0.32 in Eq. (1), or with weighted BCE (Eq. (1) without s_a), because only this stage has two fine-tuned copies of the same encoder. The two DINOv3 copies are two runs of the same recipe.

Mix of the two models	PETA	Market	Avg.	Source
Logit average, nothing learned	**82.45**	86.30	**84.38**	89.46
Learned ratio per attribute	82.43	**86.31**	84.37	89.49
Logit average plus a correction learned from				
DINOv3 attribute vectors	81.15	85.99	83.57	90.93
CLIP attribute vectors	81.55	86.06	83.81	90.52
both, concatenated into an MLP	81.27	85.92	83.60	91.02
both, ratio per attribute	81.25	85.95	83.60	90.97
both, ratio per image	81.01	85.86	83.44	91.14

Table 3. Learning the mix on top of the two separately fine-tuned models of the ensemble, which stay fixed (held-out Challenge_Avg). Corrections are computed by a linear layer and start at zero. They are trained with the loss of Eq. (1) on 85% of the validation images of the two source datasets (about 22k images in the PETA fold and 14k in the Market fold). The Source column is the two-fold mean Challenge_Avg on the remaining 15% of source images, which the fusion heads do not see in training.

whether a better mix can be learned after the two models are trained, without joint training as in UniQuery. The two models stay fixed, and only a small fusion head on top of their outputs is trained (Tab. 3). The first fusion head only learns one mixing ratio of the two logits per attribute. The other five add a correction to the logit average, computed from the attribute vectors of the query heads. An attribute vector is the vector that a query head produces for one attribute just before its classifier. In the two-fold mean, no fusion head beats the plain logit average, and the five with a correction lower Challenge_Avg by 0.6 to 0.9 points. Their failure does not come from poor learning. On source images that they did not see in training, these five beat the logit aver-

Encoder pair	Training	PETA	Market	Avg.
DINOv3 + CLIP (submitted)	fine-tuned	**82.54**	86.39	**84.46**
DINOv3 + CLIP	frozen	81.07	85.80	83.43
C-RADIOv4-H + CLIP	fine-tuned	81.87	86.43	84.15
DINOv3 + C-RADIOv4-H	fine-tuned	82.11	**86.50**	84.31
DINOv3 + SigLIP2-L	fine-tuned	81.81	86.07	83.94

Table 4. Freezing or replacing the encoders of UniQuery (held-out Challenge_Avg). With frozen encoders, only 32.6M parameters are trained.

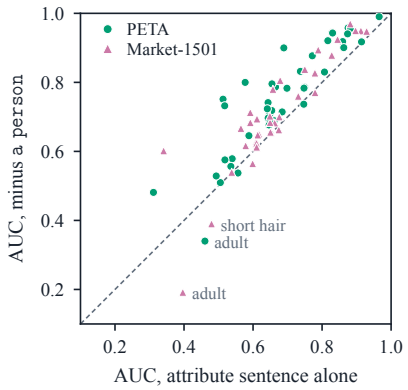


Figure 6. Zero-shot AUC of the text score on validation images, one point per attribute. Points more than 0.05 below the diagonal are named.

age by 1.1 to 1.7 points. For the fusion heads we tried, a mix learned after training therefore only fits the source datasets and does not carry over to a new dataset. This is why UniQuery learns its mix while it trains the two encoders.

Must the encoders be fine-tuned? BRAVE [14] keeps several pretrained encoders frozen and trains only a module whose learned queries read their tokens. We tried this with UniQuery by freezing both encoders and training only the three query heads and the two linear layers. UniQuery then loses 1.0 point in the mean and falls below independently fine-tuned DINOv3 (Tab. 4). The CLIP query head loses 2.7 points, while the DINOv3 query head loses only 0.6 points. In this task, the encoders therefore need fine-tuning, and freezing hurts CLIP more than DINOv3.

Does the CLIP text score help? Every attribute sentence describes a person, so the similarity of an image to an attribute sentence shares a part with its similarity to `a person`. This shared part only says how much the image looks like a person, not whether the person has the attribute. Subtracting it should keep the part that is specific to the attribute. Figure 6 tests this before training with the AUC (area under the ROC curve). The AUC is the probability

Submission	Loss	Challenge_Avg
Independent ensemble	weighted BCE†	0.7893
Independent ensemble	logit-adjusted	0.7989
UniQuery, variant	logit-adjusted	0.7992
UniQuery (submitted)	logit-adjusted	0.8001

Table 5. UPAR 2027 Track 1 hidden test set (scale 0 to 1). Final models are trained on the training and validation splits of all three datasets. All submissions use a threshold of 0.5 without post-processing, and the variant is the one in Tab. 1. †With WiSE-FT [36] for CLIP.

that an image with the attribute gets a higher score than an image without it. On PETA, the subtraction raises the mean AUC from 0.68 to 0.76 and raises the AUC for 37 of the 40 attributes, and Market-1501 shows the same direction. After training, the benefit is much smaller. Removing the text score lowers Challenge_Avg by only 0.15 points on PETA and 0.05 points on Market, about the size of the gap between two seeds. One possible reason is that once CLIP is fine-tuned on labeled data, the CLIP query head learns most of this information directly from the image tokens. The text score therefore carries correct attribute information but is a minor component, not the main source of the gain of UniQuery.

Hidden test set. On the hidden test set, both UniQuery submissions are above the independent ensemble with the same loss (Tab. 5). These gaps are only about the size of the gap between the two UniQuery submissions, so the hidden test set agrees with the held-out datasets in direction but cannot measure the size of the gain. Our final submission is UniQuery.

5. Limitations

Our results are measured on the PAR task of UPAR 2027, with its three datasets (PA100k, PETA and Market-1501) and its 40 attributes. We have not tested UniQuery on other PAR datasets, on other person-centric tasks, or with encoder pairs beyond those in this paper. We therefore cannot yet say whether mixing encoders per attribute brings the same benefit in those settings. Most results also come from a single seed.

6. Conclusion

This paper studies how to combine two foundation encoders for pedestrian attribute recognition on unseen cameras. DINOv3 and CLIP are wrong on the same image less often than two copies of one encoder, which is consistent with their different pretraining, and pairing them helps more than pairing two such copies. UniQuery fine-tunes both encoders

in one network and lets one query per attribute decide which encoder to draw on. Our main finding is that the mix needs to be learned together with the encoders, through a joint query head placed next to the query head of each encoder. Without the joint query head at inference, UniQuery no longer beats the ensemble, and the mixes we tried on top of finished models only fit the source datasets. Inside the joint query head, each attention head chooses one encoder, with lower-body and accessory attributes leaning on DINOv3 and upper-body colors on CLIP. For cross-domain PAR, this suggests letting the model assign encoders to attributes during training, rather than mixing finished models with one ratio for all attributes.

References

[1] Daniel Bolya et al. Perception encoder: The best visual embeddings are not at the output of the network. *arXiv preprint arXiv:2504.13181*, 2025.

[2] Doanh C. Bui, Thinh V. Le, and Ba Hung Ngo. C^2T-Net: Channel-aware cross-fused transformer-style networks for pedestrian attribute recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 351–358, 2024.

[3] Doanh C. Bui, Thinh V. Le, Ba Hung Ngo, and Tae Jong Choi. CLEAR: Cross-transformers with pre-trained language model for person attribute recognition and retrieval. *Pattern Recognition*, 164:111486, 2025.

[4] Yung-Sung Chuang et al. Meta CLIP 2: A worldwide scaling recipe. *arXiv preprint arXiv:2507.22062*, 2025.

[5] Yuanzheng Ci, Yizhou Wang, Meilin Chen, Shixiang Tang, Lei Bai, Feng Zhu, Rui Zhao, Fengwei Yu, Donglian Qi, and Wanli Ouyang. UniHCP: A unified model for human-centric perceptions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17840–17852, 2023.

[6] Mickael Cormier, Andreas Specker, Julio C. S. Jacques Junior, Lucas Florin, Jürgen Metzler, Thomas B. Moeslund, Kamal Nasrollahi, Sergio Escalera, and Jürgen Beyerer. UPAR challenge: Pedestrian attribute recognition and attribute-based person retrieval – dataset, design, and results. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 166–175, 2023.

[7] Mickael Cormier, Andreas Specker, Julio C. S. Jacques Junior, Lennart Moritz, Jürgen Metzler, Thomas B. Moeslund, Kamal Nasrollahi, Sergio Escalera, and Jürgen Beyerer. UPAR challenge 2024: Pedestrian attribute recognition and attribute-based person retrieval – dataset, design, and results. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 359–367, 2024.

[8] Yubin Deng, Ping Luo, Chen Change Loy, and Xiaoou Tang. Pedestrian attribute recognition at far distance. In *Proceedings of the 22nd ACM International Conference on Multimedia*, pages 789–792, 2014.

[9] Fartash Faghri, Pavan Kumar Anasosalu Vasu, Cem Koc, et al. MobileCLIP2: Improving multi-modal reinforced training. *arXiv preprint arXiv:2508.20691*, 2025.

[10] Alex Fang, Gabriel Ilharco, Mitchell Wortsman, Yuhao Wan, Vaishaal Shankar, Achal Dave, and Ludwig Schmidt. Data determines distributional robustness in contrastive language image pre-training (CLIP). In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 6216–6234, 2022.

[11] Alex Fang, Albin Madappally Jose, Amit Jain, Ludwig Schmidt, Alexander Toshev, and Vaishaal Shankar. Data filtering networks. In *International Conference on Learning Representations (ICLR)*, 2024.

[12] Enrico Fini et al. Multimodal autoregressive pre-training of large vision encoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[13] Jian Jia, Houjing Huang, Xiaotang Chen, and Kaiqi Huang. Rethinking of pedestrian attribute recognition: A reliable evaluation under zero-shot pedestrian identity setting. *arXiv preprint arXiv:2107.03576*, 2021.

[14] Oğuzhan Fatih Kar, Alessio Tonioni, Petra Poklukar, Achin Kulshrestha, Amir Zamir, and Federico Tombari. BRAVE: Broadening the visual encoding of vision-language models. In *European Conference on Computer Vision (ECCV)*, pages 113–132, 2024.

[15] Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 1994.

[16] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[17] Dangwei Li, Xiaotang Chen, and Kaiqi Huang. Multi-attribute learning for pedestrian attribute recognition in surveillance scenarios. In *Proceedings of the 3rd IAPR Asian Conference on Pattern Recognition (ACPR)*, pages 111–115, 2015.

[18] Dangwei Li, Zhang Zhang, Xiaotang Chen, and Kaiqi Huang. A richly annotated pedestrian dataset for person retrieval in real surveillance scenarios. *IEEE Transactions on Image Processing*, 28(4):1575–1590, 2019.

[19] Da Li, Zhang Zhang, Yifan Zhang, Zhen Jia, and Caifeng Shan. AdaGPAR: Generalizable pedestrian attribute recognition via test-time adaptation. *Machine Intelligence Research*, 22(4):783–796, 2025.

[20] Shilong Liu, Lei Zhang, Xiao Yang, Hang Su, and Jun Zhu. Query2Label: A simple transformer way to multi-label classification. *arXiv preprint arXiv:2107.10834*, 2021.

[21] Xihui Liu, Haiyu Zhao, Maoqing Tian, Lu Sheng, Jing Shao, Shuai Yi, Junjie Yan, and Xiaogang Wang. HydraPlus-Net: Attentive deep features for pedestrian analysis. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 350–359, 2017.

[22] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *International Conference on Learning Representations (ICLR)*, 2021.

[23] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.

[24] Mike Ranzinger, Greg Heinrich, Jan Kautz, and Pavlo Molchanov. AM-RADIO: Agglomerative vision foundation model – reduce all domains into one. In *CVPR*, 2024.

[25] Mike Ranzinger, Greg Heinrich, Collin McCarthy, Jan Kautz, Andrew Tao, Bryan Catanzaro, and Pavlo Molchanov. C-RADIOv4 (tech report). *arXiv preprint arXiv:2601.17237*, 2026.

[26] Tal Ridnik, Gilad Sharir, Avi Ben-Cohen, Emanuel Ben-Baruch, and Asaf Noy. ML-Decoder: Scalable and versatile classification head. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 32–41, 2023.

[27] Mehran Adibi Sedeh, Assia Benbihi, Romain Martin, Marianne Clausel, and Cédric Pradalier. AttriVision: Advancing generalization in pedestrian attribute recognition using CLIP. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 317–328, 2025.

[28] Oriane Siméoni et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025.

[29] Andreas Specker, Mickael Cormier, and Jürgen Beyerer. UPAR: Unified pedestrian attribute recognition and person retrieval. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 981–990, 2023.

[30] Quan Sun, Yuxin Fang, Ledell Wu, Xinlong Wang, and Yue Cao. EVA-CLIP: Improved training techniques for CLIP at scale. *arXiv preprint arXiv:2303.15389*, 2023.

[31] Shixiang Tang, Cheng Chen, Qingsong Xie, Meilin Chen, Yizhou Wang, Yuanzheng Ci, Lei Bai, Feng Zhu, Haiyang Yang, Li Yi, Rui Zhao, and Wanli Ouyang. HumanBench: Towards general human-centric perception with projector assisted pretraining. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21970–21982, 2023.

[32] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, Ziteng Wang, Rob Fergus, Yann LeCun, and Saining Xie. Cambrian-1: A fully open, vision-centric exploration of multimodal LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[33] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? exploring the visual shortcomings of multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9568–9578, 2024.

[34] Michael Tschannen et al. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025.

[35] David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[36] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo-Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7959–7971, 2022.

[37] Mitchell Wortsman et al. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 23965–23998, 2022.

[38] Hu Xu, Saining Xie, Xiaoqing Ellen Tan, Po-Yao Huang, Russell Howes, Vasu Sharma, Shang-Wen Li, Gargi Ghosh, Luke Zettlemoyer, and Christoph Feichtenhofer. Demystifying CLIP data. In *International Conference on Learning Representations (ICLR)*, 2024.

[39] Xiaokang Zhang and Hai-Miao Hu. ASRL: Correlation-robust pedestrian attribute recognition via fixed orthogonal classifier. *Neural Networks*, 196:108408, 2026.

[40] Liang Zheng, Liyue Shen, Lu Tian, Shengjin Wang, Jingdong Wang, and Qi Tian. Scalable person re-identification: A benchmark. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1116–1124, 2015.

[41] Yibo Zhou, Hai-Miao Hu, Yirong Xiang, Xiaokang Zhang, and Haotian Wu. Pedestrian attribute recognition as label-balanced multi-label learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, pages 61964–61978, 2024.

[42] Chao Zhu, Liu Yang, and Zihang Han. Cross-domain pedestrian attribute recognition: Evaluation criteria, a new baseline and remote sensor-based application. *Sensors*, 26(4): 1306, 2026.